

TALENT ACQUISITION OPERATIONS · SELECTION SCIENCE

High-volume hiring. Evidence intact.

A synthetic portfolio simulation and enterprise-oriented reference implementation under development. It models five requisitions, structured assessment, recruiter capacity, and fairness monitoring.

OpenCATS model

40 / 40 / 20 composite

100% synthetic

4,000

applicants across five
requisition families

120

modeled hires · 3.0%
conversion

28.5d

modeled mean time-to-fill

91.8%

synthetic feedback within 48
hours

Use boundary: demonstration only and not suitable for real applicant data. Operational use requires independent legal, security, accessibility, and validation review.

SYSTEM MAP

One proposed path from requisition to hire.

1

Requisition design

Document headcount, grade, manager, dates, and role-specific evidence requirements.

2

Application and knockout

The design specifies job-related rules and demographic-field isolation as production requirements.

3

Structured assessment

Synthetic work-sample and interview evidence uses anchored scoring and common criteria.

4

Governed composite

Executable tests verify 40% work sample + 40% structured interview + 20% job knowledge.

5

Modeled decision view

Synthetic capacity, turnaround, conversion, impact, and override flags remain visible together.

GOVERNED MEASURES

Metrics stay interpretable because their denominators stay visible.

Funnel

4,000 modeled applications →
986 shortlisted → 500
assessed → 120 offered and
hired.

Selection formula

The implemented 40/40/20
calculation weights job-
relevant work samples and
structured evidence.

Service level

1,836 of 2,000 synthetic
interview records classify
within the 48-hour standard.

Fairness signal

A modeled 0.87 adverse-
impact ratio is above the 0.80
monitoring threshold.

Capacity

Modeled stage volume and
turnaround illustrate where
bottlenecks could be
investigated.

Traceability

Synthetic candidate,
requisition, interview, score,
outcome, and disposition fields
remain linked.

Interpretation control: the 4/5ths rule is a screening indicator, not proof that a process is legally compliant or free from bias.

RISK AND OPERATING DISCIPLINE

Human judgment remains reviewable.

- **Halo-effect review:** a +0.68 subjective-versus-composite gap flags CAND-2026-0013 for review without diagnosing intent.
- **Structured rubrics:** documentation specifies anchored questions, common probes, and scoring notes to limit uncontrolled variance.
- **Role-based access:** proposed production RBAC separates recruiter, interviewer, manager, HRIS, and audit permissions; no runtime enforcement exists.
- **Reason codes:** the proposed workflow requires documented, job-related evidence for dispositions and overrides.
- **Privacy:** public data is synthetic. Retention, erasure, encryption, and field isolation are production requirements, not implemented controls.
- **UAT:** 14 documented scenarios specify acceptance checks; they do not prove production readiness.

Built deliverables: synthetic datasets, dictionary, workflow and RACI, rubrics, selection-evidence rationale, fairness-monitoring matrix, privacy matrix, UAT register, dashboard, and deck.

EXECUTIVE TAKEAWAY

Use evidence without confusing a model for a system.

What leaders can examine

Modeled funnel loss, requisition service levels, evidence gaps, and capacity signals.

What the artifact demonstrates

A five-requisition reference workflow, deterministic formulas, traceable records, and paginated static views.

What production teams must add

RBAC, encryption, immutable logs, retention and erasure, integrations, backups, recovery, and incident response.

What the model does not claim

No production system, real applicant data, compliance, accessibility conformance, validated fairness, or local predictive validity.

Research anchor: Sackett et al. (2022) selection-method meta-analysis, applied as an evidence foundation rather than a claim of local predictive validity.